

Zhiming (Michael) Xu

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EDUCATION

Tongji University

09/2022-07/2026

- Bachelor of Engineering in Data Science, GPA: 4.86/5.00 (Rank Top 5% in honored class, Guohao College).

RESEARCH EXPERIENCE

Robotics and Artificial Intelligence Lab, Tongji University

03/2024-03/2026

1. Zhiming Xu, Haodong Yang, Chengju Liu, Qijun Chen, Chenpeng Yao, *KinematicRL: A Sim-to-Real Reinforcement Learning Framework For Social Navigation With Kinodynamic Feasibility*. IEEE Transactions on Automation Science and Engineering (T-ASE 2026). [\[Paper\]](#)

- Developed a mathematically grounded higher-order control framework to reveal state-error decays exponentially as control order increases, and leveraged this insight to introduce a second-order action space that enforces dynamic feasibility for differential drive robots.
- Designed a maximum-entropy controller to pretrain the second-order policy via a divergence minimization objective, achieved $\sim 4\times$ faster convergence speed compared to standard mean-squared-error objective.
- Architected a novel gated transformer policy for social navigation. Incorporated unbiased residual gating and domain randomization to resolve optimization instability under varying crowd densities, increased navigation success rates by 20% and reduced frequency of invading human comfort distance to 3%.
- Engineered a full sim-to-real navigation stack featuring a 2D LiDAR perception module with SOTA velocity estimation error (0.11m/s), and deployed the system to navigate complex, real-world pedestrian environments.

2. Zhiming Xu, Weitao Zhou, Xianghui Pan, Nanshan Deng, Chengju Liu, Qijun Chen, Chenpeng Yao, *Dynamics Are Learned, Not Told: Semi-Supervised Discovery of Latent Dynamics Geometries For Zero-Shot Policy Adaptation*. International Conference on Machine Learning (ICML 2026). [\[Paper\]](#)

- Theoretically, revealed a monotonic relationship between target domain regret and the Lipschitz smoothness of a trajectory dynamics encoder, and proved the Lipschitz constant can be upper-bounded through contrastive learning.
- Practically, integrated multi-positive InfoNCE into a variational framework, translating theoretical guarantees into zero-shot reinforcement learning policy generalization.
- On MuJoCo control benchmarks, the method consistently outperformed parameter-centric baselines (e.g., Rapid Motor Adaptation) under out-of-distribution dynamics shifts, including structural failures (+29% reward on Walker2d) and time-varying dynamics (+12% reward on Ant), while also improving in-distribution performance stability.
- Qualitatively, trajectories induced by similar dynamics were adjacent in latent space, and this underlying geometry (perception-level zero-shot extrapolation) unlocks out-of-distribution generalization.

Beijing Academy of Artificial Intelligence

03/2026-Present

- Investigated end-to-end whole-body reinforcement learning for loco-manipulation with a Unitree Go2 quadruped equipped with an ARX5 arm. Demonstrated that jointly training the mobile base and manipulator within a unified policy consistently outperforms two-stage training and policies trained separately for locomotion and manipulation.
- Deployed the learned whole-body RL policy on the physical robot and developed a VR teleoperation pipeline for collection of real-world loco-manipulation demonstrations (e.g., ground-level object pickup and cup-filling tasks).
- Investigated training VLA models to output end-effector poses with the trained whole-body RL controller, where precise tracking cannot be assumed as in conventional IK-based pipelines. Used VR goal point as the learning target for VLA models, where human visual feedback compensates for the controller tracking bias.

Research Interests: Deep Reinforcement Learning, Robotics, Whole-body Control, Representation Learning.

INDUSTRY EXPERIENCE

Multimodal Agent Group, Autonomous Driving Divison, NIO Inc.

07/2025-11/2025

Project I: Learning From Multimodal Demonstrations With Auto-Regressive Architecture

- Developed a custom clustering algorithm to extract 384 representative motion tokens from 20k driving clips. Achieved

- trajectory reconstruction error of 0.06m regardless of trajectory length. Utilized top-k selection to inject controlled noise during tokenization to increase robustness of the model.
- Constructed a query-based architecture, where ego query and generated tokens attend, via deformable attention, to BEV features, then interact with context queries to autoregressively generate variable-length path.

Project II: Framework For Fine-tuning Auto-Regressive Models With Reinforcement Learning

- Integrated RL into auto-regressive planners via a custom parallel fine-tuning pipeline. Implemented PPO/GRPO/GSPO on model built with mmdet3d, reducing collision rate of baseline model by 18%.
- Designed tailored reward functions (e.g., proximity and curvature penalties) to address obstacle clearance and trajectory kinematic infeasibilities, bring key metrics (e.g., minimum distance to obstacles, trajectory curvature) into acceptable operating ranges.

Project III: Research of Vision-Language Navigation Foundation Model

- Developed a dual-system architecture referencing the *InternVLA-NI* model, featuring an asynchronous structure with a VLM (slow system) and an action expert (fast execution). Incorporated a plan token to pass VLM-generated pixel-goal information to action expert, enabling joint training of two systems.
- Constructed in-door navigation evaluation scenarios based on real autonomous driving contexts to assess scene understanding and instruction following, laying the foundation for future VLA system research and development.

ENGINEERING PROJECTS

Motion Planning With Deep Reinforcement Learning 03/2025-07/2025

- Constructed a lightweight 2D LiDAR-based navigation simulator with vectorized sensor simulation and data collection, achieving $\sim 50\times$ faster training efficiency compared to commonly used Gazebo simulator.
- Implemented classic DRL algorithms (SAC, PPO, TD3, DQN, etc) in a plug-and-play training framework.
- Added support for dynamic/static obstacles of arbitrary shapes, customizable robot geometries, domain and dynamics randomization to enhance policy robustness.
- Achieved zero-shot policy transfer to Gazebo simulator. Integrated with a global planner to form a hierarchical system combining reactive collision avoidance with long-range goal-selection in complex indoor environments.

Reinforcement Learning For Enhancing Chain-of-Thought Consistency 07/2025-08/2025

- Constructed high-quality chain-of-thought supervision data on the GQA dataset and proposed a consistency metric that detects reward hacking in Visual Question Answering (VQA) by verifying the alignment between reasoning keywords and final answers.
- Developed a two-stage training framework: (1) Supervised fine-tuning of *Qwen2.5-VL-3B-Instruct* for structured reasoning outputs; (2) RL fine-tuning (built upon *trl+vLLM*) with customized reward functions for accuracy, format compliance, reasoning-answer consistency. Compared GRPO vs. GSPO for reasoning quality improvement.
- Achieved +28.6% accuracy and +11.7% chain-of-thought consistency gains on GQA using GSPO.

SELECTED AWARDS

- National Scholarship x2 (Highest scholarship awarded by Chinese Government, top 0.1%) 09/2023, 09/2025
- Alishan Scholarship (Sole undergraduate recipient in Tongji University) 09/2024
- National First Prize in National Mathematical Contest in Modeling (Awarded to top 0.5% teams) 11/2024
- National First Prize in National English Debating Competition (National Best Speaker Award) 06/2024

SERVICES

Captain & Mentor, Tongji English Debating Team 03/2024-07/2026

- Led the university's top debating team and organized weekly sessions on motion analysis, devoted 2 hours per week mentoring a new generation of 20+ debaters in logical reasoning and public speaking.

SKILLS & OTHERS

- Programming Languages: Python, C/C++, MATLAB, JavaScript
- Robotics Related: ROS, IsaacSim/IsaacLab MuJoCo, Habitat
- Others: PyTorch, huggingface transformers/lerobot, mmdet/mmcv, vim, git, cmake